UNIT 2 CLOUD COMPUTING SERVICE MODELS AND DEPLOYMENT MODELS

2.1 Learning objectives

- Describe different service models.
- Elaborate benefits of service models and issues related to them.
- Describe different deployment models.
- Know advantages and disadvantages of deployment models.

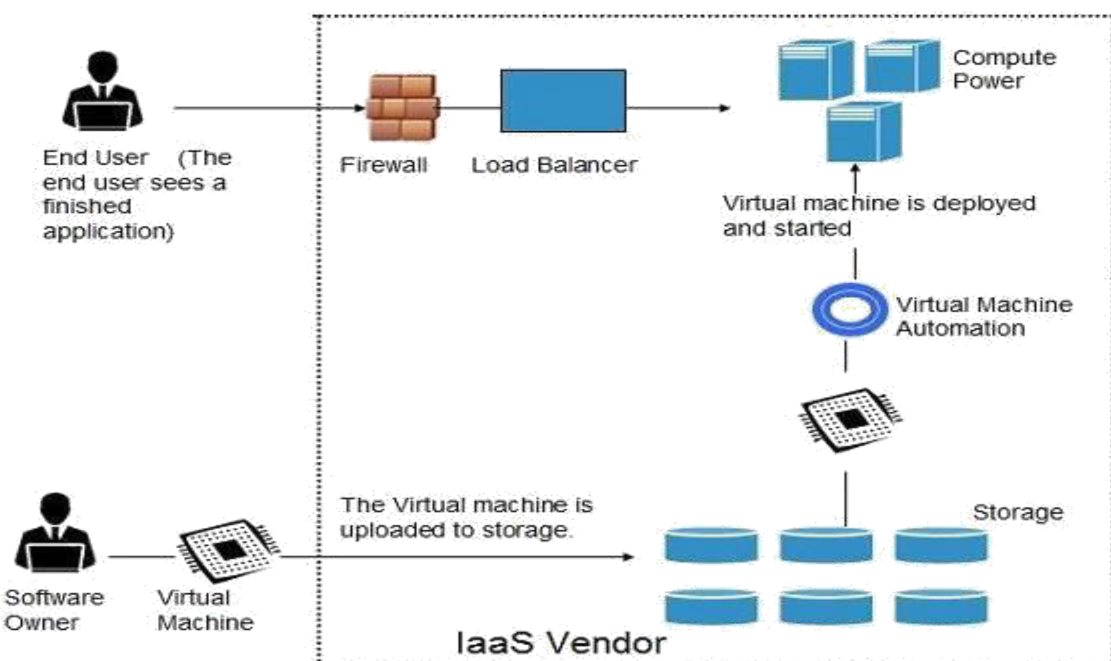
2.2. SERVICE MODELS

2.2.1. Infrastructure as a service

Iaas provider's access to fundamental resources such as physical machines, virtual machines, virtual storage, etc., Apart from these resources, the IaaS also offers:

- Virtual machine disk storage
- Virtual local area network (VLANs)
- Load balancers
- IP addresses
- Software bundles

All of the above resources are made available to end user via server virtualization. Moreover, these resources are accessed by the customers as if they own them.

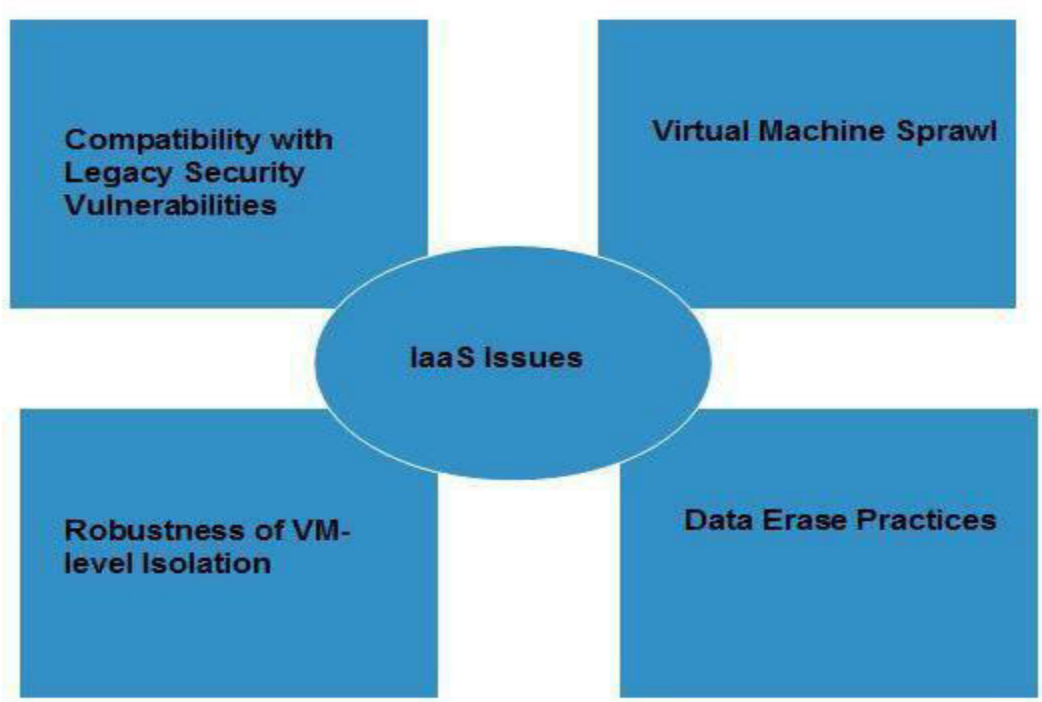


Benefits:

IaaS allows the cloud provider to freely locate the infrastructure over the Internet in a cost-effective manner. Some of the key benefits of IaaS are listed below:

- Full Control of the computing resources through Administrative Access to VMs.
- Flexible and Efficient renting of Computer Hardware.
- Portability, Interoperability with Legacy Applications.

Issues:
IaaS shares issues with PaaS and SaaS, such as Network dependence and browser based risks. It also have some specific issues associated with it. These issues are mentioned in the following diagram:



- Compatibility With Legacy Security Vulnerabilities

Because IaaS offers the consumer to run legacy software in provider's infrastructure, therefore it exposes consumers to all of the security vulnerabilities of such legacy software.

- Virtual Machine Sprawl

The VM can become out of date with respect to security updates because IaaS allows the consumer to operate the virtual machines in running, suspended and off state. However, the provider can automatically update such VMs, but this mechanism is hard and complex.

- Robustness Of Vm-Level Isolation
  IaaS offers an isolated environment to individual consumers through hypervisor. Hypervisor is a software layer that includes hardware support for virtualization to split a physical computer into multiple virtual machines.

- Data Erase Practices
  The consumer uses virtual machines that in turn use the common disk resources provided by the cloud provider. When the consumer releases the resource, the cloud provider must ensure that next consumer to rent the resource does not observe data residue from previous consumer.

Characteristics:

Here are the characteristics of IaaS service model:

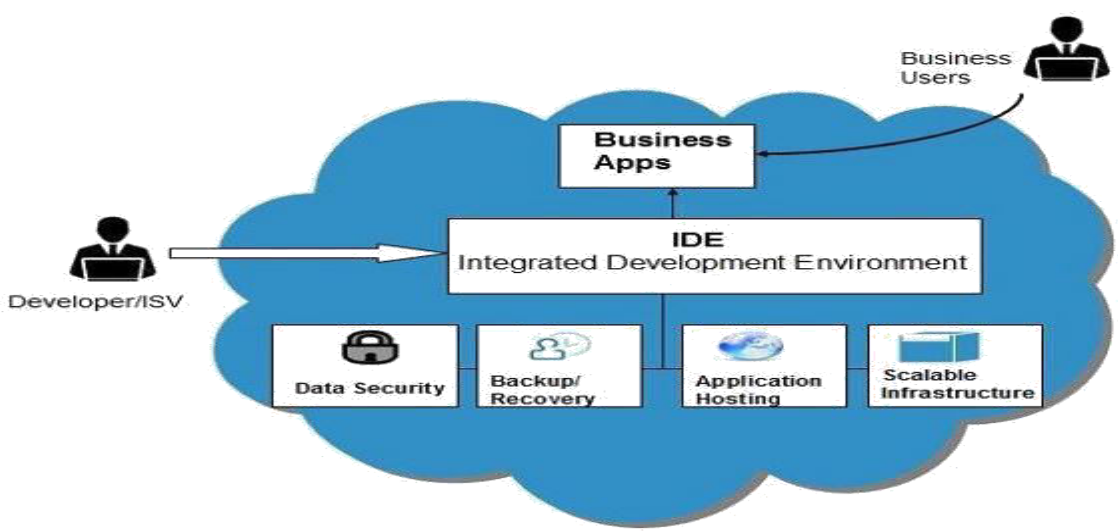
- Virtual machines with pre-installed software.
- Virtual machines with pre-installed Operating Systems such as Windows, Linux, and Solaris.
- On-demand availability of resources.
- Allows to store copies of particular data in different locations.
- The computing resources can be easily scaled up and down.

2.2.2 Platform as a Service(PAAS)-

It also offers development & deployment tools, required to develop applications. PaaS has a feature of point-and-click tools that enables non-developers to create web applications.Google's App Engine, Force.com are examples of PaaS offering vendors. Developer may log on to thesewebsites and use the built-in API to create web-based applications.

But the disadvantage of using PaaS is that the developer lock-in with a particular vendor. For example, an application written in Python against Google's API using Google's App Engine is likely to work only in that environment. Therefore, the vendor lock-in is the biggest problem in PaaS.

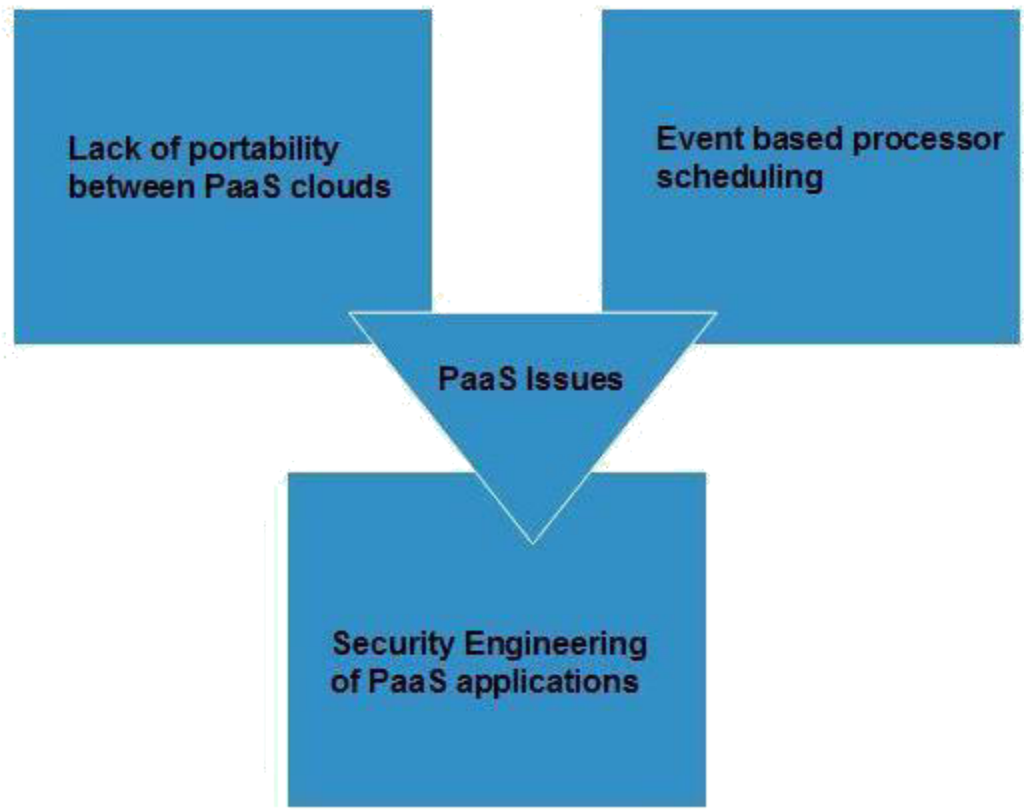
The following diagram shows how PaaS offers an API and development tools to the developers and how it helps the end user to access business applications.



Benefits:

- Lower Administrative Overhead
  Consumer need not to bother much about the administration because it's the responsibility of cloud provider.
- Lower Total Cost Of Ownership
  Consumer need not purchase expensive hardware, servers, power and data storage.
- Scalable Solutions
  It is very easy to scale up or down automatically based on application resource demands.
- More Current System Softwar
  It is the responsibility of the cloud provider to maintain software versions and patch installations.

Issues:



- Lack Of Portability Between Paas Clouds
  Although standard languages are used yet the implementations of platforms services may vary. For example, file, queue, or hash table interfaces of one platform may differ from another, making it difficult to transfer workloads from one platform to another.
- Event Based Processor Scheduling

The PaaS applications are event oriented which poses resource constraints on applications, i.e., they have to answer a request in a given interval of time.

- Security Engineering Of Paas Applications
  Since the PaaS applications are dependent on network, PaaS applications must explicitly use cryptography and manage security exposures.

Characteristics:

Here are the characteristics of PaaS service model:

- PaaS offers browser based development environment. It allows the developer to create database and edit the application code either via Application Programming Interface or point-and-click tools.
- PaaS provides built-in security, scalability, and web service interfaces.
- PaaS provides built-in tools for defining workflow and approval processes and defining business rules.
- It is easy to integrate with other applications on the same platform.
- PaaS also provides web services interfaces that allow us to connect the applications outside the platform.

2.2.3 Software as a Service(SaaS)

This model allows providing software application as a service to the end users. It refers to a software that is deployed on a hosted service and is accessible via Internet. There are several

SaaS applications, some of them are listed below:

- Billing and Invoicing System
- Customer Relationship Management (CRM) applications
- Help Desk Applications
- Human Resource (HR) Solutions

Some of the SaaS applications are not customizable such as an Office Suite. But SaaS provides us Application Programming Interface (API), which allows the developer to develop a customized application.

Characteristics:

Here are the characteristics of SaaS service model:

- SaaS makes the software available over the Internet.
- The Software are maintained by the vendor rather than where they are running.
- The license to the software may be subscription based or usage based. And it is billed on recurring basis.
- SaaS applications are cost effective since they do not require any maintenance at end user side.
- They are available on demand.

- They can be scaled up or down on demand.
- They are automatically upgraded and updated.
- SaaS offers share data model. Therefore, multiple users can share single instance of infrastructure. It is not required to hard code the functionality for individual users.
- All users are running same version of the software.

Benefits:

Using SaaS has proved to be beneficial in terms of scalability, efficiency, performance and much more. Some of the benefits are listed below:

- Modest Software Tools
- Efficient use of Software Licenses
- Centralized Management & Data
- Platform responsibilities managed by provider
- Multitenant solutions

Issues:

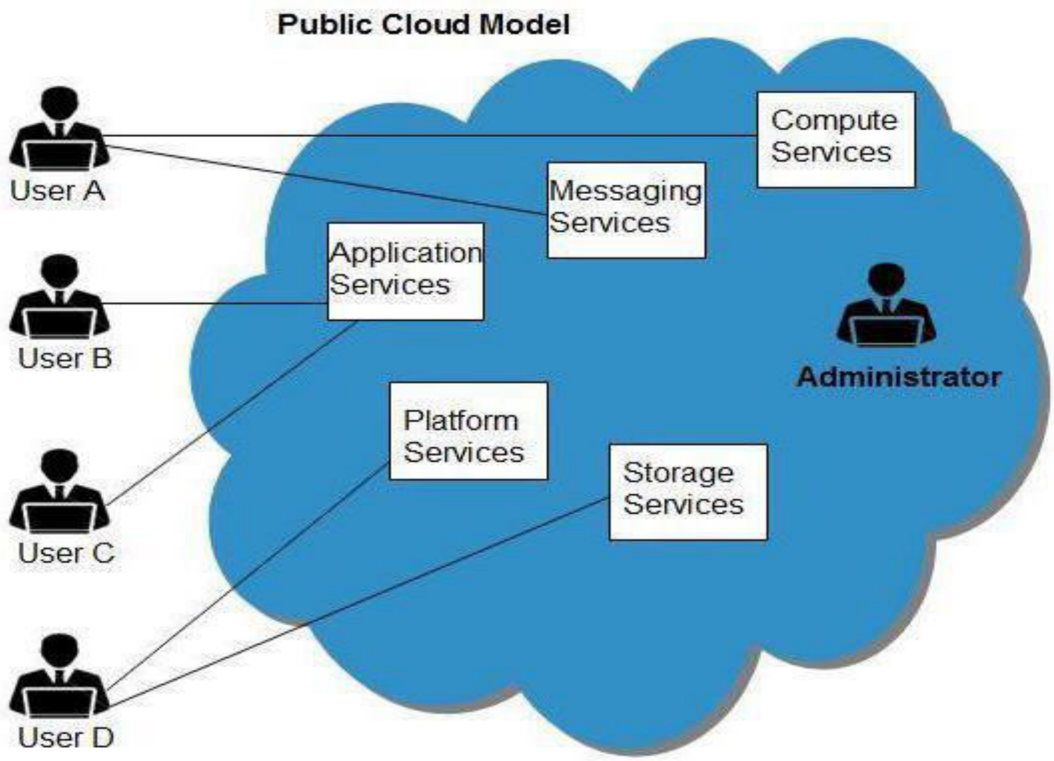
There are several issues associated with SaaS, some of them are listed below:

- Browser Based Risks
  If the consumer visits malicious website and browser becomes infected, and the subsequent access to SaaS application might compromise the consumer's data. To avoid such risks, the consumer can use multiple browsers and dedicate a specific browser to access SaaS applications or can use virtual desktop while accessing the SaaS applications.
- Network Dependence
  The SaaS application can be delivered only when network is continuously available. Also network should be reliable but the network reliability cannot be guaranteed either by cloud provider or the consumer.
- Lack Of Portability Between Saas Clouds
  Transferring workloads from one SaaS cloud to another is not so easy because work flow, business logics, user interfaces, support scripts can be provider specific

2.3 Deployment Models

2.3.1 Public clouds

The Public Cloud allows systems and services to be easily accessible to general public, e.g., Google, Amazon, Microsoft offers cloud services via Internet.



Benefits:

- Cost Effective
  Since public cloud share same resources with large number of consumer, it has low cost.
- Reliability:
  Since public cloud employs large number of resources from different locations, if any of the resource fail, public cloud can employ another one.
- Flexibility
  It is also very easy to integrate public cloud with private cloud, hence gives consumers a flexible approach.
- Location Independence
  Since, public cloud services are delivered through Internet, therefore ensures location independence.
- Utility Style Costing
  Public cloud is also based on pay-per-use model and resources are accessible whenever consumer needs it.
- High Scalability
  Cloud resources are made available on demand from a pool of resources, i.e., they can be scaled up or down according the requirement.

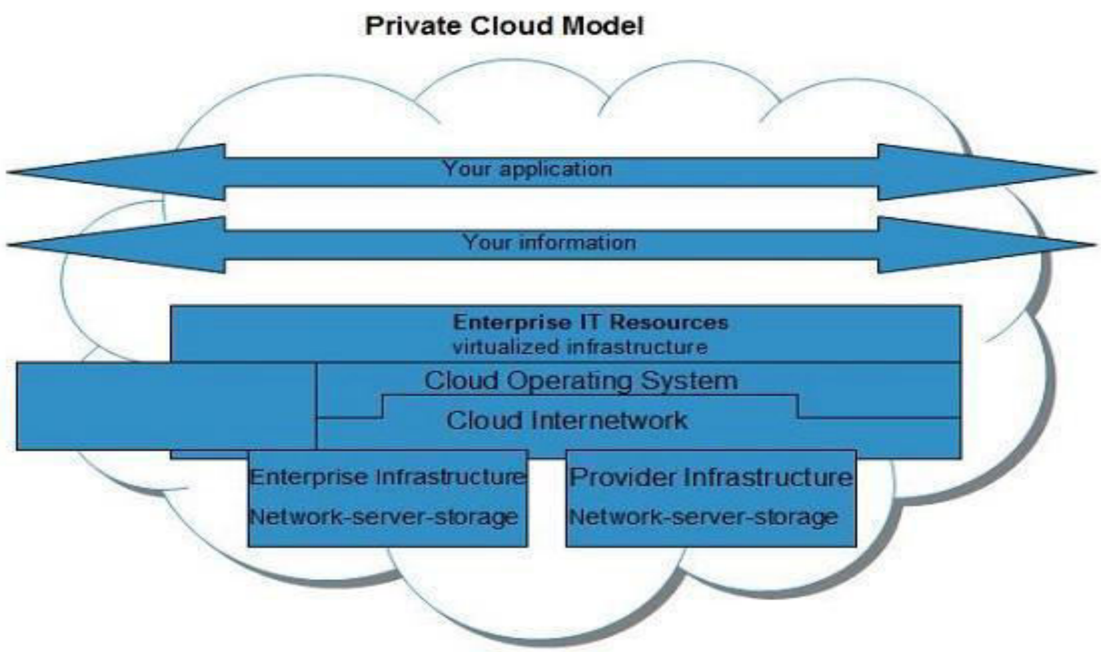
Disadvantages:

Here are the disadvantages of public cloud model:

- Low Security
  In public cloud model, data is hosted off-site and resources are shared publicly, therefore does not ensure higher level of security.
- Less Customizable
  It is comparatively less customizable than private cloud.

2.3.2 Private cloud

The Private Cloud allows systems and services to be accessible within an organization. The Private Cloud is operated only within a single organization. However, it may be managed internally or by third-party.



Benefits:

- Higher Security And Privacy
  Private cloud operations are not available to general public and resources are shared from distinct pool of resources, therefore, ensures high security and privacy.
- More Control
  Private clouds have more control on its resources and hardware than public cloud because it is accessed only within an organization.
- Cost And Energy Efficiency
  Private cloud resources are not as cost effective as public clouds but they offer more efficiency than public cloud.

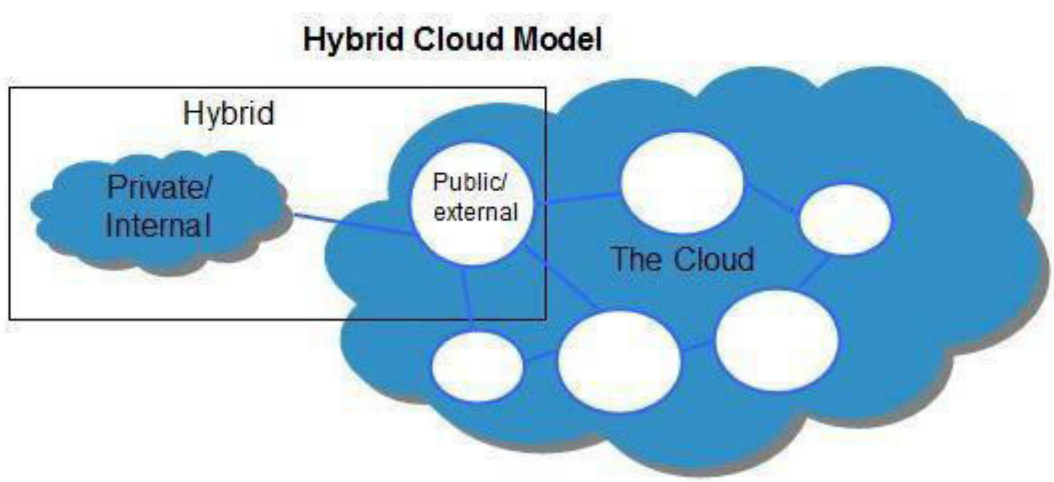
Disadvantages:

Here are the disadvantages of using private cloud model:

- Restricted Area
  Private cloud is only accessible locally and is very difficult to deploy globally.
- Inflexible Pricing
  In order to fulfill demand, purchasing new hardware is very costly.
- Limited Scalability
  Private cloud can be scaled only within capacity of internal hosted resources.

2.3.3 Hybrid cloud

The Hybrid Cloud is a mixture of public and private cloud. Non-critical activities are performed using public cloud while the critical activities are performed using private cloud.



Benefits:

- Scalability
  It offers both features of public cloud scalability and private cloud scalability.
- Flexibility
  It offers both secure resources and scalable public resources.
- Cost Efficiencies

Public cloud are more cost effective than private, therefore hybrid cloud can have this saving.

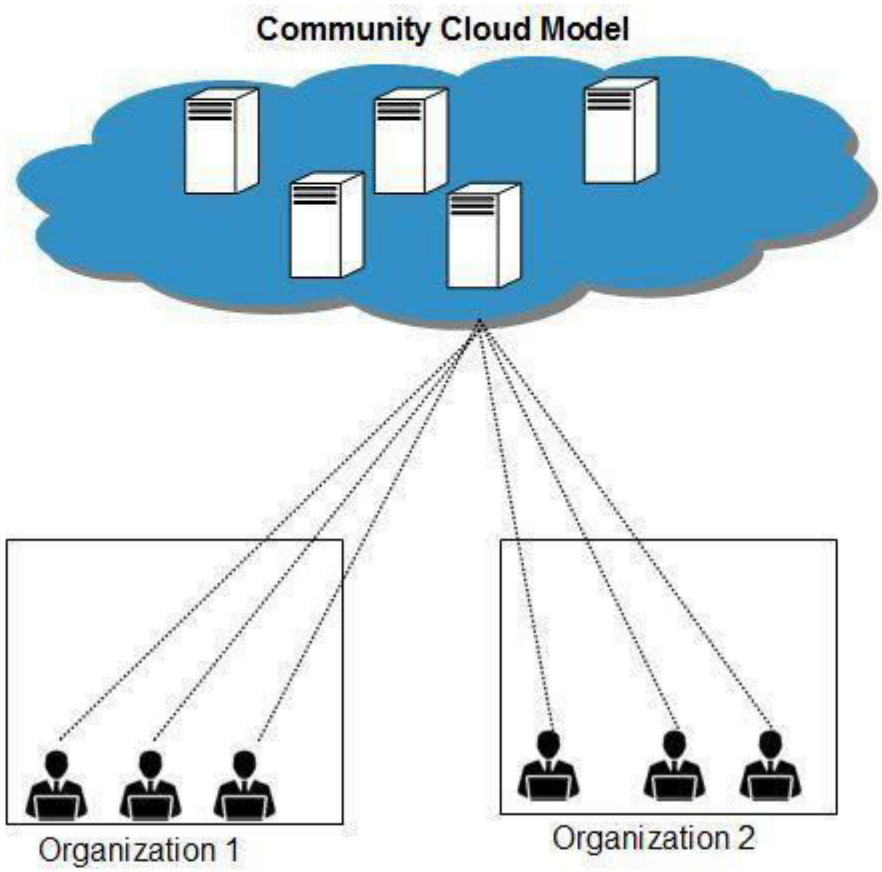
- Security
  Private cloud in hybrid cloud ensures higher degree of security.

Disadvantages:

- Networking Issues
  Networking becomes complex due to presence of private and public cloud.
- Security Compliance
  It is necessary to ensure that cloud services are compliant with organization's security policies.

2.3.4 Community cloud

The community Cloud allows system and services to be accessible by group of organizations. It shares the infrastructure between several organizations from a specific community. It may be managed internally or by the third-party.



Benefits:

There are many benefits of deploying cloud as community cloud model. The following diagram shows some of those benefits:

- Cost Effective
  Community cloud offers same advantage as that of private cloud at low cost.
- Sharing Between Organizations
  Community cloud provides an infrastructure to share cloud resources and capabilities among several organizations.
- Security
  Community cloud is comparatively more secure than the public cloud.

Issues:

- Since all data is housed at one location, one must be careful in storing data in community cloud because it might be accessible by others.
- It is also challenging to allocate responsibilities of governance, security and cost.

Very short questions

1. What are different service models? Name them.
2. Name fundamental resources of Iaas?
3. Give two benefits of Iaas?
4. Paas is also known as.
5. What is private cloud?
6. What is hybrid cloud?
7. What is community cloud?

Short questions

1. Name different cloud service models with suitable example?
2. Explain characteristics of Iaas model?
3. Define private cloud along with diagram.
4. Explain benefits of private cloud?
5. What are the issues found with Paas?

Long questions

1. Explain Iaas along with four benefits, issues and characteristics.
2. What is the difference between public cloud, private cloud, hybrid cloud, community cloud?
3. Explain cloud computing deployment models in detail.

UNIT 3 VIRTUALIZATION

3.1 Learning objectives

- Know about virtualization.
- Describe hypervisor.
- Elaborate types of hypervisor

3.2 Virtualization

Virtualization is a technique, which allows sharing single physical instance of an application or resource among multiple organizations or tenants (customers). It does so by assigning a logical name to a physical resource and providing a pointer to that physical resource when demanded.

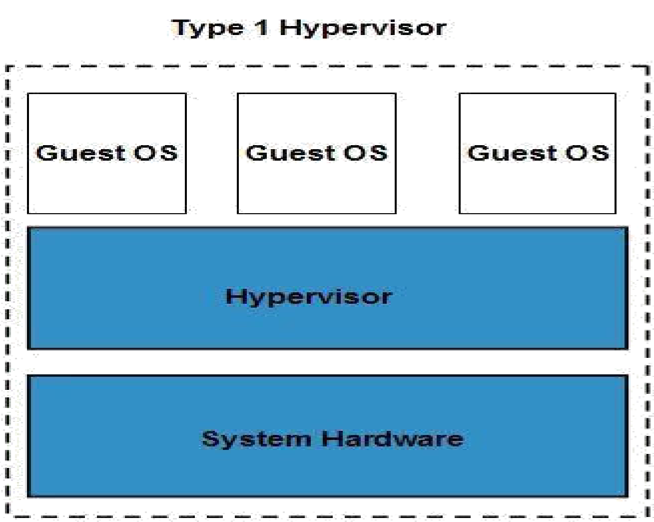
3.2.1 Virtualization Concept

Creating a virtual machine over existing operating system and hardware is referred as Hardware Virtualization. Virtual Machines provide an environment that is logically separated from the underlying hardware. The machine on which the virtual machine is created is known as host machine and virtual machine is referred as a guest machine. This virtual machine is managed by a software or firmware, which is known as hypervisor.

3.2.2 Hypervisor

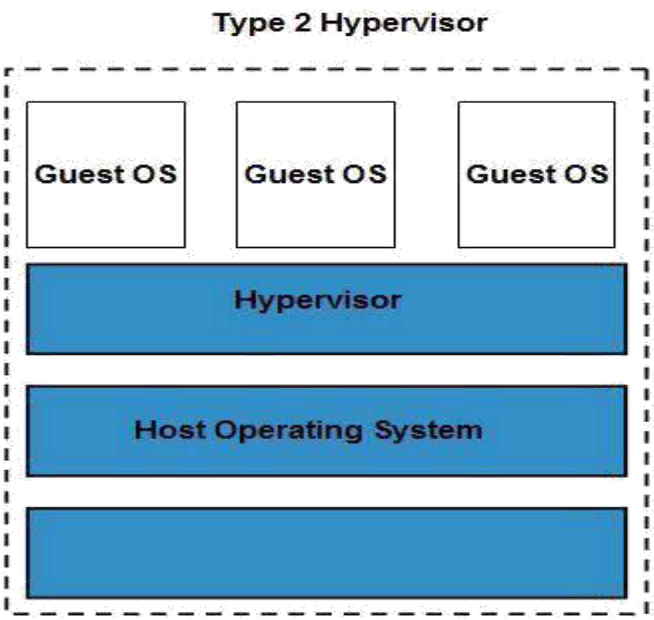
Hypervisor is a firmware or low-level program that acts as a Virtual Machine Manager. There are two types of hypervisor:

Type 1 hypervisor runs on bare system. LynxSecure, RTS Hypervisor, Oracle VM, Sun xVM Server, Virtual Logic VLX are examples of Type 1 hypervisor. The following diagram shows the Type 1 hypervisor.



The type1 hypervisor does not have any host operating system because they are installed on a bare system. Type 2 hypervisor is a software interface that emulates the devices with which a system normally interacts. Containers, KVM, Microsoft Hyper V, VMWare Fusion, Virtual

Server 2005 R2, Windows VirtualPC and VMWare workstation 6.0 are examples of Type 2 hypervisor. The following diagram shows the Type 2hypervisor



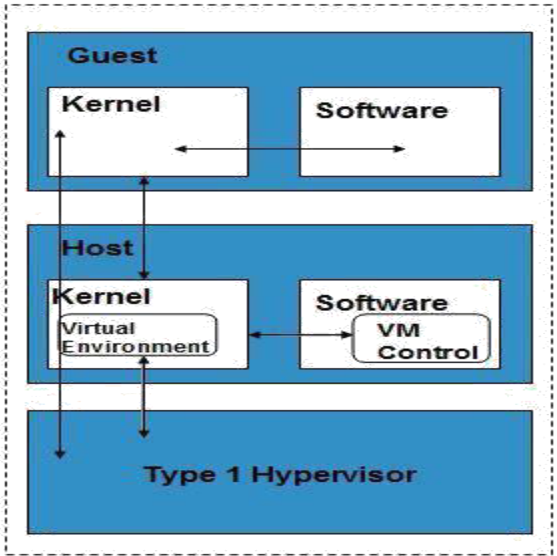
3.3 Types of Hardware Virtualization

Here are the three types of hardware virtualization:

1. Full Virtualization
2. Emulation Virtualization
3. Para virtualization

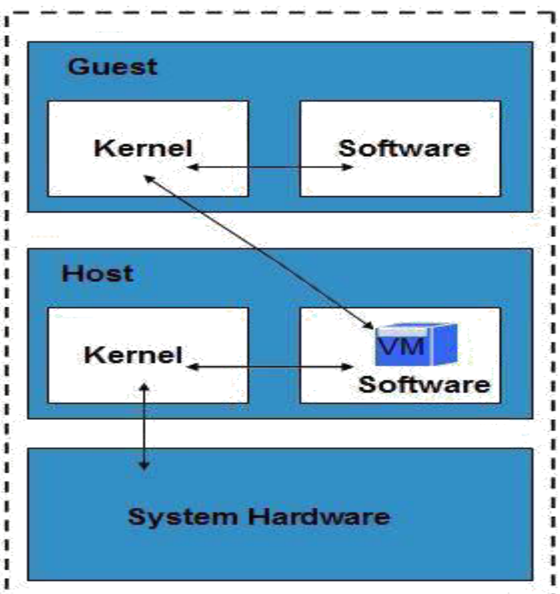
3.3.1 FULL VIRTUALIZATION

In Full Virtualization, the underlying hardware is completely simulated. Guest software does not require any modification to run.



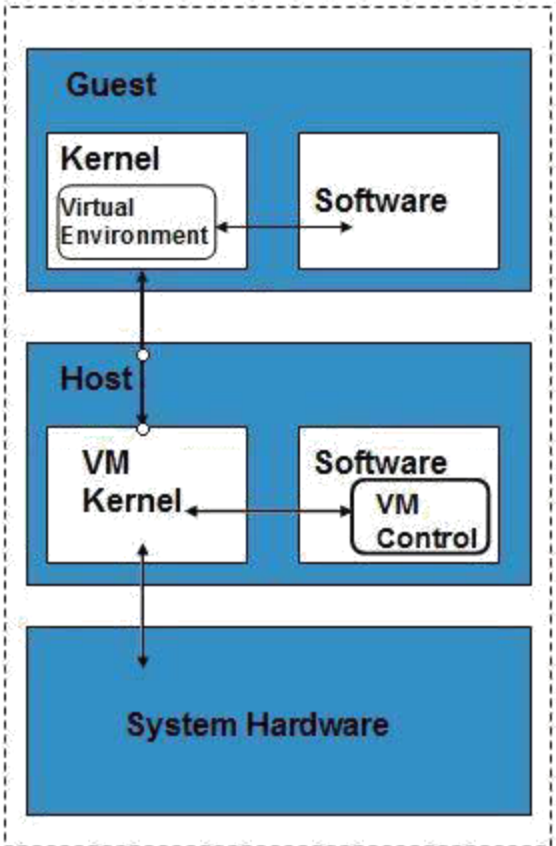
3.3.2 EMULATION VIRTUALIZATION

In Emulation, the virtual machine simulates the hardware and hence become independent of the it. In this, the guest operating system does not require modification.



3.3.3 PARAVIRTUALIZATION

In Para virtualization, the hardware is not simulated. The guest software run their own isolated domains



VMware vSphere is highly developed infrastructure that offers a management infrastructure framework for virtualization. It virtualizes the system, storage and networking hardware.

Very short questions

1. What is virtualization?
2. Name type of virtualization?
3. Define full virtualization?
4. What is hypervisor?
5. Name the types of hypervisor?

Short questions

1. Define full virtualization along with diagram?
2. Define emulation virtualization along with diagram?
3. What is type 1 hypervisor?
4. What is para virtualization?

Long questions

1. What are types of hypervisor?
2. What is virtualization and its types?

UNIT 4 Computing Technologies

4.1 Learning Objectives

- Know about cluster computing, its types, and applications.
- Describe utility computing, its properties.
- Elaborate ubiquitous computing, its applications

4.2 Cluster Computing

4.2 .1 What is a Cluster?

In its most basic form, a cluster is a system comprising two or more computers or systems (called nodes) which work together to execute applications or perform other tasks, so that users who use them, have the impression that only a single system responds to them, thus creating an illusion of a single resource (virtual machine). This concept is called transparency of the system. As key features for the construction of these platforms is included elevation : reliability, load balancing and performance.

4.2.2 Types of cluster

4.2.2.1 High Availability

High Availability Cluster aims to maintain the availability of services provided by a computer system by replicating servers and services through redundant hardware and software reconfiguration. Several computers are acting together as one, each one monitoring the others and taking their services if any of them will fail. The complexity of the system must be software that should bother to monitor other machines on a network, know what services are running, those who are running, and what to do in case of a failure. Loss in performance or processing power are usually acceptable, the main goal is not to stop. There are some exceptions, like real-time and mission critical systems.

4.2.2.2 Load balancing

When we do load balancing between servers that have the same ability to respond to a client, we started having problems because one or more servers can respond to requests made and communication is impaired. So we put the element that will make balancing between servers and users, and configure it to do so, however we can put multiple servers on one side that, for the customers, they appear to be only one address. A classic example would be the Linux Virtual Server, or simply prepare a DNS load balancer. The element of balance will have an address, where customers try to make contact, called Virtual Server (VS), which redirects traffic to a server in the server pool. This element should be software dedicated to doing all this management, or may be a network device that combines hardware performance and software to make the passage of the packages and load balancing in a single device.

4.2.3 Applications of cluster computing

It has applications in making movies.

This technology is useful as commercial servers.

Clusters are being used as replicated storage and backup servers that provide the essential fault tolerance and reliability for critical applications. Cluster computing has application in internet search engine, efficient search services. There are many commercial cluster products designed for distributed databases and web servers. Cluster computing can be used for load balancing and high performance. It is used as a reliability low cost form of parallel processing machine for scientific and other applications that lend themselves to parallel operations.

4.2.4 Advantages of cluster computing

- With clustering performance of computer highly increases.
- Capacity to perform large amount of task
- High availability.
- Incremental growth.

4.2.5 Disadvantages of cluster computing

- It is difficult to develop software for distributed system.
- Risk of data security.
- Difficulty in Fault finding.
- Difficult to handle by layman.

4.3 Utility computing

The term Utility Computing refers to utility computing technologies and business models, which provide a service provider to its customers IT services, and they charge you by consumption. Examples of such IT services are computing power, storage or applications. As a service provider to the computer center of a company comes into question, in which case the customer would be the individual divisions of the company.
The term utility refers to utility services such as electricity, telephone, water and gas that are provided by a utility company. Similar to the electricity or telephone if the customer receives the utility computing, the computing power on a shared computer network, its consumption is measured and billed on that basis.

Examples of computational resources
Computation time (CPU time) is the leading resource along with the memory usage. Not limited to physical equipment, the files and network connections, and other resources as well as virtual memory space are also considered computer resources.

- CPU time
- Physical memory and virtual memory
- Hard drive space and access time
- Network bandwidth
- Environment variable, etc. are also some of the computer resources

4.3.1 Properties of utility computing
Although there are many different definitions of utility computing, it will normally include the following five characteristics of utility computing.

- Scalability

The utility computing must be ensured that under all conditions sufficient IT resources are available. Increasing the demand for a service may, its quality (e.g., response time) does not suffer.

- Demand pricing

So far, companies have to buy his own hardware and software when they need computing power. This IT infrastructure must be paid in advance of the rule, regardless of the intensity with which the company uses them later. Technology vendors to achieve this link, for example, the fact that the lease rate for their servers depends on how many CPUs has enabled the customer. If it can be measured in a company as much computing power to claim the individual sections in fact, may be the IT costs in internal cost directly attributable to the individual departments. Other forms of connection with the use of IT costs are possible.

Standardized Utility Computing Services

The utility computing service provider offers its customers a catalog of standardized services. These may have different service level agreements (Agreement on the quality and the price of an IT) services. The customer has no influence on the underlying technologies such as the server platform.

- Utility Computing and Virtualization

To share the web and other resources in the shared pool of machines can be used virtualization technologies. This will divide the network into logical resource instead of the physical resources available. An application is assigned no specific pre-determined servers or storage of any but a free server runtime or memory from the pool.

- Automation

Repetitive management tasks such as setting up a new server or the installation of updates can be automated. Moreover, automatically allocate resources to services and the management of IT services to be optimized, with service level agreements and operating costs of IT resources must be considered.

4.3.2 Advantages of Utility Computing

Utility computing reduces the cost of IT, given that existing resources can be used more effectively. Moreover, the costs are transparent and the various departments of a company can be directly assigned. In the IT departments will be fewer people needed for operational activities.

The companies achieve greater flexibility, because their IT resources more quickly and easily adapt to fluctuating demand. Overall, it is easier to manage the entire IT structure, as there will no longer be made for each application, which is a benefit for specific IT infrastructure

4.4 Ubiquitous computing

Pervasive computing, also called ubiquitous computing, is the growing trend of embedding computational capability (generally in the form of microprocessors) into everyday objects to make them effectively communicate and perform useful tasks in a way that minimizes the end user's need to interact with computers as computers. Pervasive computing devices are network-connected and constantly available.

The goal of pervasive computing is to make devices "smart," thus creating a sensor network capable of collecting, processing and sending data, and, ultimately, communicating as a means to adapt to the data's context and activity; in essence, a network that can understand its surroundings and improve the human experience and quality of life.

Often considered the successor to mobile computing, ubiquitous computing and, subsequently, pervasive computing, generally involve wireless communication and networking technologies, mobile devices, embedded systems, wearable computers, RFID tags, middleware and software agents. Internet capabilities, voice recognition and artificial intelligence are often also included.

Pervasive computing applications can cover energy, military, safety, consumer, healthcare, production and logistics.

An example of pervasive computing is an Apple Watch informing a user of a phone call and allowing him to complete the call through the watch. Or, when a registered user for Amazon's streaming music service asks her Echo device to play a song, and the song is played without any other user intervention.

4.4.1 Properties of ubiquitous computing

- Immerse computers in a real environment.
- Sensors support internet with and control the environment.
- Limited power supply, storage memory and bandwidth.
- Operate unattended.
- Devices are mobile/wireless.

4.4.2 Applications of ubiquitous computing

- Communications: as a cross-application, the communications area affects all forms of exchange and transmission of data, information, and knowledge. Communications thus represents a precondition for all information technology domains
- Logistics: tracking logistical goods along the entire transport chain of raw materials, semi-finished articles, and finished products (including their eventual disposal) closes the gap in IT control systems between the physical flow and the information flow. This offers opportunities for optimizing and automating logistics that are already apparent today.
- Motor traffic: automobiles already contain several assistance systems that support the driver invisibly. Networking vehicles with each other and with surrounding telematics systems is anticipated for the future.
- Military: the military sector requires the provision of information on averting and fighting external threats that is as close-meshed, multidimensional, and interrelated as possible. This comprises the collection and processing of information. It also includes the development of new weapons systems.
- Production: in the smart factory, the flow and processing of components within manufacturing are controlled by the components and by the processing and transport stations themselves. Ubiquitous computing will facilitate a decentralized production system that will independently configure, control and monitor itself.
- Smart homes: in smart homes, a large number of home technology devices such as heating, lighting, and ventilation and communication equipment become smart objects that automatically adjust to the needs of the resident.
- E-commerce: the smart objects of ubiquitous computing allow for new business models with a variety of digital services to be implemented. These include location-based services, a shift from selling products to renting them, and software agents that will

instruct components in ubiquitous computing to initiate and carry out services and business transactions independently.

- Inner security: identification systems, such as electronic passport and the already abundant smart cards, are applications of ubiquitous computing in inner security. In the future, monitoring systems will become increasingly important—for instance, in protecting the environment or surveillance of key infrastructure such as airports and the power grid.
- Medical technology: Increasingly autarkic, multifunctional, miniaturized and networked medical applications in ubiquitous computing offer a wide range of possibilities for monitoring the health of the elderly in their own homes, as well as for intelligent implants.

Very short questions

1. Define cluster?
2. What is cluster computing?
3. Define load balancing?
4. What are high available cluster?
5. Define ubiquitous computing?

Short questions

1. What are the advantages of cluster computing?
2. What are the types of cluster computing?
3. What is utility computing?
4. What role ubiquitous computing has in smart homes and military?
5. What are the properties of ubiquitous computing?

Long questions

1. What is cluster computing? What are its applications, its advantages and disadvantages?
2. What are the applications of ubiquitous computing?
3. What is utility computing, its properties, its advantages?
4. Describe various computing Technologies?